MATH 251: TOPOLOGY II

SPRING 2026 PRACTICE PROBLEMS

Contents

Note: All citations are to Munkres's textbook, *Topology*, 2nd Edition, unless indicated otherwise. When a problem statement has a proof in Munkres, try your best to find your own proof, before comparing with his.

For any integer $n > 0$, the topology on $\mathbf{R}^n$ is the analytic topology unless otherwise specified.

0. Review of Topology I

Problem 1. Show that no two of the spaces

$$(0,1), \quad (0,1], \quad [0,1]$$

are homeomorphic. *Hint:* What happens if you remove any point from $(0,1)$?

Problem 2. Suppose that $A \subseteq X$. Recall that the *interior* of A in X is

$$\mathrm{Int}_X(A) := \{x \in X \mid x \in U \text{ for some } U \text{ open in } X \text{ such that } U \subseteq A\}$$

and the *closure* of A in X is

$$\mathrm{Cl}_X(A) := \{x \in X \mid x \in K \text{ for all } K \text{ closed in } X \text{ such that } K \supseteq A\}.$$

Show the identity $\mathrm{Cl}_X(A) = X \setminus \mathrm{Int}_X(X \setminus A)$.

Problem 3. Suppose that $A \subseteq Y \subseteq X$, where Y is a subspace of X.

(1) Show that $\mathrm{Int}_Y(A) \supseteq \mathrm{Int}_X(A)$.
(2) Give an example where $\mathrm{Int}_Y(A) \neq \mathrm{Int}_X(A)$. *Hint:* You can assume $Y = A$.

Problem 4. Let X be the real line $\mathbf{R}$ in its finite complement, or *cofinite*, topology. Show that every sequence of points in X converges to every point of X simultaneously. Deduce that X is not Hausdorff.

Problem 5. Show that any metric space is Hausdorff.

Problem 6. Show that for any integer $n \geq 1$, the analytic topology on $\mathbf{R}^n$ matches the product topology on $\mathbf{R} \times \cdots \times \mathbf{R}$, where there are n factors.

Problem 7. Let $p\colon \mathbf{R} \to S^1$ be the map

$$p(t) = (\cos(2\pi t), \sin(2\pi t)).$$

For any $a, b \in \mathbf{R}$, we define the *(open) arc* $J_{a,b} \subseteq S^1$ to be

$$J_{a,b} = \{p(t) \mid a < t < b\}.$$

Show that the collection of arcs $\{J_{a,b} \mid a, b \in \mathbf{R}\}$ satisfies the definition of a basis (Munkres page 78).

Problem 8. Show that the following topologies on S^1 are all the same:

- The topology generated by the basis $\{J_{a,b} \mid a, b \in \mathbf{R}\}$ in Problem 7.
- The subspace topology that S^1 inherits from its inclusion into $\mathbf{R}^2$.
- The quotient topology that S^1 inherits from the surjective map $p\colon \mathbf{R} \to S^1$.

Problem 9. Let $f_1, f_2, f_3\colon A \to X$ be continuous maps. Suppose that φ is a homotopy from f_1 to f_2 and ψ is a homotopy from f_2 to f_3. Construct a homotopy from f_1 to f_3 explicitly in terms of φ and ψ.

Problem 10. In the setup of Problem 9, suppose that

$$\begin{aligned} A &= [0,1], \\ f_1(0) &= f_2(0) = f_3(0), \\ f_1(1) &= f_2(1) = f_3(1). \end{aligned}$$

Show that in this case, if φ and ψ are path homotopies, then we can choose the solution of (1) to be a path homotopy as well.

Problem 11. Let $f, g\colon X \to Y$ and $F, G\colon Y \to Z$ be continuous. Show that

$$\text{if } f \sim g \text{ and } F \sim G, \quad \text{then } F \circ f \sim G \circ g.$$

Problem 12. Use the $f = g$ case of Problem 11 to show: If X, Y are nonempty and Y is contractible, then any continuous map from X into Y is *nulhomotopic*: that is, homotopic to a constant map.

Problem 13. Give an explicit homotopy equivalence between S^1 and $\mathbf{R}^2 \setminus \{(0,0)\}$, and show that it is indeed a homotopy equivalence.

Problem 14. Suppose that:

- $f\colon X \to Y$ and $g\colon Y \to X$ define a homotopy equivalence between X and Y.
- $j\colon Y \to Z$ and $k\colon Z \to Y$ define a homotopy equivalence between Y and Z.

Show that $j \circ f\colon X \to Z$ and $g \circ k\colon Z \to X$ define a homotopy equivalence between X and Z.

Hint: Among other things, you will need to show that $g \circ k \circ j \circ f \sim \mathrm{id}_X$. But we are given that $k \circ j \sim \mathrm{id}_Y$. Use Problem 11 to show that $k \circ j \sim \mathrm{id}_Y$ implies $g \circ k \circ j \circ f \sim \mathrm{id}_X$.

Problem 15. Show that if the identity map of a space X is homotopic to a constant map with value $c \in X$, then X is homotopy equivalent to $\{c\}$. Deduce that all *contractible* spaces are homotopy equivalent to each other.

Problem 16. A subset $A \subseteq \mathbf{R}^n$ is *star convex* if and only if there is some point $c \in A$ such that the line segment between c and any other point of A is contained within A. That is, for any $a \in A$ and $t \in [0,1]$, we have $(1-t)c + ta \in A$.

(1) Show that if A is star convex, then A is contractible.
(2) Show that if A is star convex, then any loop in A based at a_0 is path homotopic to the constant loop.
(3) Give a star convex subset of $\mathbf{R}^2$ that is not convex.

Exam 1 Topics

Exam 1 will be a **closed-book** exam. It will be held in-class, 12:40–2:00 pm, on ~~January 26~~ ~~February 2~~ February 4.

You should know by heart the definitions of the following terms. You should be able to state multiple examples of each (with justification), and in some cases, non-examples.

- Munkres §13. Bases
- §18. Continuous maps
- §15. The product topology (only on finite direct products)
- §16. Subspaces
- §22. Quotient spaces
- §17. Interiors and closures
- §23–24. Separations, connectedness
- §23. Paths, path connectedness
- §26–27 Compactness
- §51. Homotopies, path homotopies
- §58. Homotopy equivalences

1. Fundamental Groups and Covering Spaces

Problem 17. Show that if X is path connected, then X is connected.

Problem 18. We say that X is *totally disconnected* if and only if its only nonempty connected subspaces are one-point sets.

(1) Show that if X is discrete, then X is totally disconnected.
(2) Show that the set of rational numbers $\mathbf{Q}$, as a subspace of (analytic) $\mathbf{R}$, is totally disconnected, but not discrete.
(3) Show that if X is totally disconnected, then $\pi_1(X, x)$ is trivial for all $x \in X$. *Hint:* Use Problem 17.

Problem 19. Consider the points $p = (1, 0)$ and $q = (-1, 0)$ in $\mathbf{R}^2$.

(1) Give two paths in S^1 from p to q that are not path homotopic.
(2) Explain why the paths in (1) become path homotopic when viewed as paths in $\mathbf{R}^2$.

Problem 20. Let $x_0, x_1 \in X$. Let $\alpha\colon [0, 1] \to X$ be a path from x_0 to x_1, and let $\bar{\alpha}$ be the reverse path defined by $\bar{\alpha}(s) = \alpha(1 - s)$.

(1) Show that $\alpha * \bar{\alpha}$ is path homotopic to e_{x_0}.
(2) Using part (1), show that the map

$$\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1) \quad \text{given by } \hat{\alpha}([\gamma]) = [\bar{\alpha} * \gamma * \alpha]$$

is a group homomorphism.
(3) Show that $\hat{\bar{\alpha}}$ is a two-sided inverse of $\hat{\alpha}$. Deduce that $\hat{\alpha}$ is an isomorphism.

Problem 21. Let $f\colon X \to Y$ be a continuous map.

(1) Show that if $\beta, \gamma\colon [0, 1] \to X$ are paths such that $\beta(1) = \gamma(0)$, then

$$f \circ (\beta * \gamma) = (f \circ \beta) * (f \circ \gamma).$$

(2) Let $x \in X$, Using part (1), show that the map

$$f_*\colon \pi_1(X, x) \to \pi_1(Y, f(x)) \quad \text{given by } f_*([\gamma]) = [f \circ \gamma]$$

is a group homomorphism.
(3) Let $g\colon Y \to Z$ be another continuous map. Show that $(g \circ f)_* = g_* \circ f_*$ as homomorphisms from $\pi_1(X, x)$ into $\pi_1(Z, g(f(x)))$.

Problem 22. Let X be a subspace of some space X', and let $f\colon X \to Y$ be a continuous map. Suppose that

$$f = f'|_X \quad \text{for some continuous map } f'\colon X' \to Y,$$

and that X' is *simply connected*, meaning X' is path connected and its fundamental group is trivial. Show that in this case, the homomorphism $f_*\colon \pi_1(X, x) \to \pi_1(Y, f(x))$ is *trivial* for all $x \in X$, meaning it sends every element of $\pi_1(X, x)$ to the identity element of $\pi_1(Y, f(x))$. *Hint:* Use part (3) of Problem 21.

Problem 23. Let X_1, X_2 be spaces with points $x_1 \in X_1$ and $x_2 \in X_2$. Let $X = X_1 \times X_2$ and $x = (x_1, x_2) \in X$. For $i = 1, 2$, let $\mathrm{pr}_i \colon X \to X_i$ be the appropriate *projection map* (Munkres page 87).

(1) Show that the map

$$\varphi\colon \pi_1(X, x) \to \pi_1(X_1, x_1) \times \pi_1(X_2, x_2) \quad \text{given by } \varphi([\gamma]) = ([\mathrm{pr}_1 \circ \gamma], [\mathrm{pr}_2 \circ \gamma])$$

is a homomorphism.

(2) Show that the map

$$\psi\colon \pi_1(X_1, x_1) \times \pi_1(X_2, x_2) \to \pi_1(X, x) \quad \text{given by } \psi([\gamma_1], [\gamma_2]) = [\gamma_1 \times \gamma_2]$$

is a two-sided inverse to φ. Deduce that

$$\pi_1(X, x) \simeq \pi_1(X_1, x_1) \times \pi_1(X_2, x_2).$$

(3) Using the conclusion to part (2), show that $\mathbf{Z}^n$ is the fundamental group of an explicit (path-connected) topological space, for any integer $n > 0$.

Problem 24. By considering what happens on fundamental groups when a point is removed, show that $\mathbf{R}^2$ and $\mathbf{R}^n$ are not homeomorphic if $n > 2$.

Problem 25. Learn the definitions of a *free group* and of a *presentation of a group (by generators and relations)*. (They are covered in Munkres §69, but you may find other sources online helpful as well.)

Let F_2 denote a free group on generators a and b. Show that the homomorphism

$$\varphi\colon F_2 \to \mathbf{Z}^2 \quad \text{given by } \varphi(a) = (1, 0) \text{ and } \varphi(b) = (0, 1)$$

is surjective and that its kernel contains the *commutator* of a and b: that is, the element $aba^{-1}b^{-1}$.

(In fact, the kernel is exactly the smallest normal subgroup of F_2 containing the commutator. This shows that $\mathbf{Z}^2$ has a presentation with generators a, b and a single relation $aba^{-1}b^{-1}$.)

Problem 26. Suppose that A, U_1, U_2 are path-connected open subsets of X such that $X = U_1 \cup U_2$ and $A = U_1 \cap U_2$. For $k = 1, 2$, let $i_k \colon A \to U_k$ and $j_k \colon U_k \to X$ be the appropriate inclusion maps. Fix a point $x \in A$.

(1) Describe a homomorphism

$$\Psi\colon \pi_1(U_1, x) \star \pi_1(U_2, x) \to \pi_1(X, x)$$

such that $\Psi|_{\pi_1(U_k, x)} = (j_k)_*$ for $k = 1, 2$. (It is necessarily unique.)

(2) Show that for any $[\gamma] \in \pi_1(A, x)$, the element

$$(i_1)_*([\gamma]) * (i_2)_*([\gamma])^{-1} \in \pi_1(U_1, x) \star \pi_1(U_2, x)$$

belongs to the kernel of Ψ. *Hint:* Compare $j_1 \circ i_1$ and $j_2 \circ i_2$.

Problem 27. Recall that $X \sqcup Y$ denotes the disjoint union of X and Y. The *wedge sum* of X and Y at points $x \in X$ and $y \in Y$ is a quotient space

$$X \vee Y = (X \sqcup Y)/\sim,$$

where $\sim$ is an equivalence relation that only identifies two distinct points when those points are x and y. Use Seifert-van Kampen to show that $\pi_1(S^1 \vee S^1, p) \simeq \mathbf{Z} \star \mathbf{Z}$ for any basepoint p.

Problem 28. Keep the hypotheses of Problem 26. Thus, the Seifert–van Kampen theorem applies to the homomorphism Ψ in part (1) of that problem. Give examples where:

(1) A is simply-connected, but X, U_1, U_2 are not. *Hint:* Problem 27.
(2) X, U_1, U_2 are simply-connected, but A is not. *Hint:* Munkres §59.
(3) X and U_1 are simply-connected, but U_2 is not.
(4) X is simply-connected, but U_1 and U_2 are not.

Problem 29. For each of the following spaces, the fundamental group is either trivial, $\mathbf{Z}$, or the free group $F_2 \simeq \mathbf{Z} \star \mathbf{Z}$. Determine which option is the case. You do not need to give explicit homeomorphisms or homotopy equivalences, but give informal descriptions (or pictures) to support your reasoning.

(1) $D \times S^1$, where $D = \{(x, y) \in \mathbf{R}^2 \mid x^2 + y^2 < 1\}$.
(2) The *torus* $S^1 \times S^1$.
(3) The bounded cylinder $S^1 \times [0, 1]$.
(4) The unbounded cylinder $S^1 \times \mathbf{R}$.
(5) The punctured plane $\mathbf{R}^2 - \{p\}$, where p is any point.
(6) *Harder:* The punctured torus $S^1 \times S^1 - \{q\}$, where q is any point.

Problem 30. Determine the fundamental group of the punctured 3-dimensional space $\mathbf{R}^3 - \{(0, 0, 0)\}$. Note that it is path-connected, so different basepoints give isomorphic fundamental groups.

Hint: Is it homotopy equivalent to a space whose fundamental group you already know?

Problem 31. Let $p\colon E \to B$ be a covering map. Show that if B is connected and $p^{-1}(b_0)$ has k elements for some $b_0 \in B$, then $p^{-1}(b)$ has k elements for all $b \in B$. In this case, we say that p is a *k-fold covering*.

Problem 32. Draw every 2-fold covering space of $S^1 \vee S^1$ up to homeomorphism. You do not need to prove that your list is exhaustive.

Note 1: Some covering spaces could be disconnected! *Note 2:* $S^1 \vee S^1$ has a symmetry of order two. If two covering spaces differ by a lift of this symmetry, you do not need to draw both.

Problem 33. Let $\mathbf{R}_+$ be the set of positive numbers. Let $p\colon \mathbf{R} \times \mathbf{R}_+ \to \mathbf{R}^2 - \{(0, 0)\}$ be defined by $p(u, r) = (r \cos(2\pi u), r \sin(2\pi u))$. This is a covering map. Find liftings along p of the following paths in $\mathbf{R}^2 - \{(0,0)\}$:

$$\begin{aligned} f(t) &= (2-t, 0), \\ g(t) &= ((1+t)\cos(2\pi t), (1+t)\sin(2\pi t)), \\ h &= f * g. \end{aligned}$$

Sketch (the images of) these paths and their liftings.

Problem 34. Let $p\colon \mathbf{R} \to S^1$ be defined by

$$p(t) = (\cos(2\pi u), \sin(2\pi u)).$$

Consider the path in $S^1 \times S^1$ given by

$$f(t) = ((\cos(2\pi t), \sin(2\pi t)), (\cos(4\pi t), \sin(4\pi t))).$$

Find an explicit lifting $\tilde{f}$ of f along $(p,p) : \mathbf{R} \times \mathbf{R} \to S^1 \times S^1$, and sketch it.

Problem 35. Find all *connected* covering spaces of S^1 up to homeomorphism.

Exam 2 Topics

Exam 2 will be an **open-book** exam. It will be held in-class, 12:40–2:00 pm, on March 2.

"Open book" means: You will be allowed to look at Munkres, *Topology*, 2nd Ed., and any notes on paper that you wrote prior to the exam, but you will not be allowed to use electronic devices, including phones, computers, tablets, *etc.*

Topics Included.

- Definition of the fundamental group $\pi_1(X, x)$
- Whether $\pi_1(X, x)$ and $\pi_1(X, x')$ are related, when x, x' belong to the same path component of X or to different path components of X
- Definition of $f_*\colon \pi_1(X, x) \to \pi_1(Y, f(x))$, for a (continuous) map $f\colon X \to Y$
- Relation between $(g \circ f)_*$ and $g_* \circ f_*$, for maps $f\colon X \to Y$ and $g\colon X \to Y$
- Relation between $\pi_1(X, x)$ and $\pi_1(Y, f(x))$, when $f\colon X \to Y$ forms part of a homotopy equivalence
- $\pi_1(X, x)$ when $X = \mathbf{R}^N$ or S^N, for different values of N
- $\pi_1(X \times Y, (x, y))$ in terms of $\pi_1(X, x)$ and $\pi_1(Y, y)$
- Explicit loops representing the elements of $\pi_1(S^1, o)$, where $o = (1, 0)$
- Surjectivity statement in the Seifert–van Kampen theorem: relation between $\pi_1(X, x)$ and $\pi_1(U_1, x) \star \pi_1(U_2, x)$ when $X = U_1 \cup U_2$ for some open U_1, U_2 such that $U_1, U_2, U_1 \cap U_2$ are each path connected and $x \in U_1 \cap U_2$
- The formula for $\pi_1(S^1 \vee S^1)$ (via Seifert–van Kampen)
- Variants of Problems 28 and 29 above
- Definition of covering maps
- Why $p(t) = (\cos(2\pi t), \sin(2\pi t))$ defines a covering map $p\colon \mathbf{R} \to S^1$, but not if we replace $\mathbf{R}$ by $[0, 1)$ or $[0, 1]$ or $(0, 1)$, *etc.*
- Some examples of coverings of $S^1 \vee S^1$ and of $S^1 \times S^1$
- The path lifting and path-homotopy lifting properties of covering maps (*e.g.*, lifting a path in X along a covering $E \to X$ to a starting point in E)
- Relation between $\pi_1(X, p(e))$ and $\pi_1(E, e)$, when $p\colon E \to X$ is a covering

Topics Not Included.

- The precise definition of a free product of groups
- The kernel statement in the Seifert–van Kampen theorem
- The Jordan separation/curve theorems

2. Fixed Points and Separations in the Plane

Problem 36. Recall that a *simple closed curve* in a space X is a subspace of X homeomorphic to S^1. Find simple closed curves $C, C' \subseteq S^1 \times S^1$ such that:

(1) $S^1 \times S^1 - C$ is path connected.
(2) $S^1 \times S^1 - C'$ is not path connected.

Hint: It may help to view $S^1 \times S^1$ as a quotient space of the square $[0,1] \times [0,1]$ in which opposite edges have been identified.

Problem 37. Find a 2×2 matrix with strictly positive real entries that does not have any positive real eigenvalues.

Problem 38. Let $f\colon S^1 \to S^1$ be continuous and nulhomotopic. Show that:

(1) There is some point $p \in S^1$ such that $f(p) = p$: that is, a *fixed point* of f.
(2) There is some point $q \in S^1$ such that $f(q) = -q$.

Hint: Theorem 55.5, or the version proved in class.

Problem 39. Recall that if $s\colon A \to X$ and $r\colon X \to A$ satisfy $r \circ s = \mathrm{id}_A$, then s is called a *section* and r is called a *retraction*.

(1) Show that in this situation,

$$\begin{aligned} &s \text{ is injective,} \\ &r \text{ is surjective,} \\ &s_*\colon \pi_1(A,a) \to \pi_1(X, s(a)) \text{ is injective for any } a \in A, \\ &r_*\colon \pi_1(X,x) \to \pi_1(A, r(x)) \text{ is surjective for any } x \in X. \end{aligned}$$

(2) Deduce from (1) that there is no retraction from the closed disk

$$B^2 := \{(x,y) \in \mathbf{R}^2 \mid x^2 + y^2 \leq 1\}$$

onto its boundary circle S^1.

(3) Deduce from (1) that if $S \subseteq \mathbf{R}^2$ is nonempty, then there is no retraction from $\mathbf{R}^2$ onto $\mathbf{R}^2 - S$. *Hint:* Show by induction on the size of S that the fundamental group of $\mathbf{R}^2 - S$ must be nontrivial.

Problem 40. Fix an integer $n \geq 0$. Writing $\|{-}\|$ to mean Euclidean distance, let

$$\begin{aligned} S^n &= \{v \in \mathbf{R}^{n+1} \mid \|v\| = 1\}, && \text{the } \textit{n-sphere}, \\ B^{n+1} &= \{v \in \mathbf{R}^{n+1} \mid \|v\| \leq 1\}, && \text{the } \textit{closed n-ball}. \end{aligned}$$

Let $\sim$ be the equivalence relation on $S^n \times [0,1]$ that collapses the subset $S^n \times \{1\}$ to a point: that is, $(v,t) \sim (v',t)$ if and only if either $t = 1$ or $v = v'$.

Write down an explicit homeomorphism from $(S^n \times [0,1])/{\sim}$ onto B^{n+1}. You should be able to show that your example is continuous and bijective, but you do not need to show that it is a homeomorphism. *Hint:* Use a generalization of the map in the "(1) $\Rightarrow$ (2)" step of Munkres's proof of Lemma 55.3.

Problem 41. Using Problem 40, show that if a continuous map $f\colon S^n \to X$ is nulhomotopic, then $f = F|_{S^n}$ for some map $F\colon B^{n+1} \to X$. *Hint:* Again, look at the proof of Lemma 55.3.

Problem 42. Assume the fact that there is no retraction from B^{n+1} onto S^n, in the sense of Problem 39. Using Problem 41, show that the identity map of S^n cannot be nulhomotopic.

Problem 43. Suppose that $f\colon X \to X$ is continuous. Show that:

(1) If X is the closed interval $[0, 1]$, then f has some fixed point. *Hint:* The Intermediate Value Theorem.
(2) If X is the half-closed interval $[0, 1)$, then f need not have fixed points.

3. Free Groups and Free Products

Problem 44. Recall that for any set X, we write F_X to denote the *free group* generated by X. Show that X and F_X are never in bijection. *Hint:* Imitate Cantor's diagonalization argument.

Problem 45. Show that the following conditions on a subset S of a group G are equivalent:

(1) The only subgroup of G containing S is G itself.
(2) The homomorphism from F_S into G induced by the inclusion map from S into G is surjective.

In this case, we say that S *generates* G, or that S is a *generating set* for G. If $S = \{s_1, s_2, \ldots\}$, then we also say that $s_1, s_2, \ldots$ *generate* G.

Problem 46. Let a, m be arbitrary positive integers. Recall that "the *residue class* of a mod m" refers to the coset $a + m\mathbf{Z}$.

(1) For which a and m does a mod m generate $\mathbf{Z}/m\mathbf{Z}$?
(2) In general, describe the subgroup generated by a mod m.

We say that a group is *cyclic* when it can be generated by a single element.

Problem 47. For any positive integer m, let

$$U_m = \{a \bmod m \mid ab \equiv 1 \bmod m \text{ for some } b\} \subseteq \mathbf{Z}/m\mathbf{Z}.$$

(1) Show that U_m forms a group under the operation of multiplication mod m.
(2) Compute U_2, U_3, U_4, U_5, U_6.
(3) Which of the groups in (2) are trivial? Which are cyclic?

Problem 48. Which subsets of $\mathbf{Z}$ are generating sets for $\mathbf{Z}$ as a group? *Hint:* Look up Bézout's Lemma.

Problem 49. Look up the definition of the *Cayley graph* of a group G equipped with a generating set S. Draw the Cayley graph in the case where:

(1) $G = \mathbf{Z}$ (under addition) and $S = \{1\}$.
(2) $G = \mathbf{Z}$ and $S = \{2, 3\}$.
(3) $G = \mathbf{Z}/5\mathbf{Z}$ and $S = \{2\}$.
(4) $G = F_S$ and $S = \{a, b, c\}$.

Problem 50. Let $(a, b), (c, d) \in \mathbf{Z}^2$ and

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

(1) Show that M^{-1} exists and has only integer entries if and only if $\det M = \pm 1$.
(2) Deduce from (1) that $(1, 0)$ and $(0, 1)$ are integer linear combinations of (a, b) and (c, d) if and only if $\det M = \pm 1$.
(3) Deduce from (2) that $\{(a, b), (c, d)\}$ generates $\mathbf{Z}^2$ if and only if $\det M = \pm 1$.

Problem 51. Keep the setup of Problem 50. If $\det M = 0$, then what can you say about the subgroup of $\mathbf{Z}^2$ generated by (a, b) and (c, d)?

Problem 52. Let a, b be arbitrary positive integers. Show that $\langle s \mid s^a, s^b \rangle$ is trivial if and only if a and b are relatively prime. *Hint:* Recall how we showed that $\langle s \mid s^b \rangle \simeq \mathbf{Z}/b\mathbf{Z}$.

Problem 53. Recall that the *order* of an element $g \in G$ is the smallest positive integer m such that $g^m = e$, if it exists, and infinity otherwise.

Show that if G is abelian, then its elements of finite order form a subgroup. It is called the *torsion subgroup* of G. If it is trivial, then we say that G is *torsion-free*.

Problem 54. Let G be a group in which $g^2 = e$ for all $g \in G$. Show that G is abelian. (Note that in this case, G is equal to its torsion subgroup.)

Problem 55. Let G be an arbitrary group. Show that:

(1) As a set, G is in bijection with the set of homomorphisms from $\mathbf{Z}$ into G. *Hint:* $\mathbf{Z}$ is isomorphic to a free group on one element.
(2) The set $\{g \in G \mid g^2 = e\}$ is in bijection with the set of homomorphisms from $\mathbf{Z}/2\mathbf{Z}$ into G.

Note: If G is not abelian, then $\{g \in G \mid g^2 = e\}$ need not be a subgroup of G!

In Problems 56 and 57 below, let G_1, G_2 be groups with presentations

$$G_i = \langle s \in S_i \mid r \in R_i \rangle \quad \text{for } i = 1, 2.$$

Recall that their *free product* $G_1 * G_2$ has the presentation

$$G_1 * G_2 = \langle s \in S_1 \sqcup S_2 \mid r \in R_1 \sqcup R_2 \rangle.$$

You may assume the following fact: If $w \in G_1 * G_2$ is a reduced word that involves elements of both G_1 and G_2, then $w \neq e$. For a reminder on the definition of a *reduced word*, see Munkres 415–416.

Problem 56. (1) Give a surjective homomorphism from $G_1 * G_2$ onto $G_1 \times G_2$. *Hint:* In class, we discussed the case where $G_1, G_2 \simeq \mathbf{Z}$.
(2) Show that if G_1 and G_2 are nontrivial, then the homomorphism in (1) is not injective.

Problem 57. For all $g \in G_1 * G_2$, we define the *length* of g to be its length as a reduced word in the elements of G_1 and G_2.

(1) Show that if x has even length at least 2, then it cannot have finite order.
(2) Show that if x has odd length at least 3, then it is conjugate to some element of shorter length. Here, recall that the *conjugates* of x are the elements gxg^{-1}, where $g \in G$.
(3) Deduce that any element of G of finite order is conjugate to an element of G_1 or G_2 of finite order.

Problem 58. Suppose that $X = X_1 \cup X_2$, where X_1, X_2 are closed, path-connected, and intersect in a single point p. (Recall that this is an example of a *wedge sum*.) Suppose that for $i = 1, 2$, the subspace X_i contains an open neighborhood of p, say W_i, such that $\{p\}$ is a deformation retract of W_i. Use Seifert–Van Kampen to show that in this situation,

$$\pi_1(X, p) \simeq \pi_1(X_1, p) * \pi_1(X_2, p),$$

Hint: In class, we discussed the case where $X_1, X_2 \simeq S^1$.

Hint: Set $U_1 = X_1 \cup W_2$ and $U_2 = X_2 \cup W_1$. Check that X_i is homotopy equivalent to U_i for $i = 1, 2$, and that $\{p\}$ is homotopy equivalent to $U_1 \cap U_2$. Then check the hypotheses needed for Seifert–Van Kampen. You may assume without proof that path-connectedness is preserved by homotopy equivalence.

Problems 59 and 60 below discuss the *real projective plane* P^2 from Munkres §60, page 372, and the *Klein bottle* K from Munkres §75, page 454.

Problem 59. Let X_1, X_2 be two copies of P^2. Let X be the quotient space formed from $X_1 \sqcup X_2$ by excising small open disks from X_1 and X_2, then gluing along the boundaries of the resulting holes.

(1) Show that X is homeomorphic to K, using the descriptions of P^2 and K as quotients of $[0, 1]^2$.
(2) Compute $\pi_1(X, x)$ by applying Seifert–van Kampen to open neighborhoods of X_1 and X_2 within X. Do not simply cite the formula for the fundamental group of a Klein bottle.

Problem 60. (1) Find a 2-fold covering $p\colon S^1 \times S^1 \to K$.
(2) Describe the homomorphism of fundamental groups induced by p.

Exam 3 Topics

Exam 3 will be an **open-book** exam. It will be held in-class, 12:40–2:00 pm, on ~~April 13~~ April 17.

"Open book" means: You will be allowed to look at Munkres, *Topology*, 2nd Ed., and any notes on paper that you wrote prior to the exam, but you will not be allowed to use electronic devices, including phones, computers, tablets, *etc.*

Topics Included.

- Sections, retractions, and their properties (Problem 39)
- Why there is no retraction from D^2 onto its subspace S^1
- Equivalent ways to describe when a map $S^1 \to X$ extends to a map $D^2 \to X$ (Munkres Lemma 55.3)
- Statement of the fixed-point theorem for D^2
- Statement of the Jordan separation theorem (for S^2 or $\mathbf{R}^2$)
- Examples of normal subgroups, homomorphisms, kernels, quotient groups
- Definition of the free group F_S on a set S, in terms of words
- How a map from a set S into a group G defines a homomorphism from F_S into G
- Examples and non-examples of generating sets for groups that we discussed, like $\mathbf{Z}$ or $\mathbf{Z}/m\mathbf{Z}$ or $\mathbf{Z}^2$
- Meaning of a group presentation $\langle s \in S \mid r \in R \rangle$, where $R \subseteq F_S$
- Statement of Seifert–van Kampen in terms of group presentations
- Presentations for the fundamental groups of the torus, Klein bottle, real projective plane, m-fold dunce cap, and how to describe elements in these groups using the presentations

Topics Not Included.

- The proofs of the Brouwer fixed-point theorem, Jordan separation theorem, or Seifert–van Kampen theorem
- The Jordan curve theorem
- Listing all possible generating sets of a given group
- Computing the precise smallest normal subgroup of F_S containing a given set R

4. Simplicial Complexes and Surfaces

Recall that if S is a finite set of points in general position in some real vector space, then we write $\Delta(S)$ for the simplex formed by its convex hull.

Problem 61. In $\mathbf{R}^3$, let $e_1 = (1,0,0)$ and $e_2 = (0,1,0)$ and $e_3 = (0,0,1)$.

(1) Determine every possible simplicial complex $\{\Delta(S)\}_S$ in which each set S is a subset of $\{e_1, e_2, e_3\}$. There should be 12, including the empty simplicial complex.
(2) Which simplicial complexes in (1) have a geometric realization that is itself a simplex?

Problem 62. For each space X below, draw or describe a geometric realization of a simplicial complex that is homeomorphic (not just homotopy equivalent) to X.

(1) S^1
(2) $S^1 \vee S^1 \vee S^1$
(3) D^2 (the set $\{(x,y) \in \mathbf{R}^2 \mid x^2 + y^2 \leq 1\}$)
(4) $S^1 \times [0,1]$
(5) S^2
(6) $S^1 \times S^1$
(7) $D^2 \times S^1$
(8) $\{(x,y,z) \in \mathbf{R}^3 \mid x^2 + y^2 + z^2 \leq 1\}$

Hint: To handle $S^1 \times S^1$ and $D^2 \times S^1$, first describe them as quotient spaces of $[0,1] \times [0,1]$ and $D^2 \times [0,1]$, respectively.

(Recall that a *triangulation* of X is a homeomorphism from X onto such a geometric realization. Each space above has a triangulation.)

Problem 63. For each space in Problem 62, compute its Euler characteristic using the triangulation you found.

Problem 64. Let $I = [0,1]$. The (closed) *Möbius band* is the quotient space

$$M = (I \times I)/{\sim},$$

where $(0,y) \sim (1, 1-y)$ for all y, and no other pairs of distinct points of $I \times I$ get identified under $\sim$.

Recall that the circle S^1 is homeomorphic to a similar quotient space. Give an explicit homotopy equivalence between M and S^1. (Hence, they have the same fundamental groups.)

Problem 65. Keep the setup of Problem 64.

(1) What space do you get if you cut M along the image of the line segment in $I \times I$ between $(0, \frac{1}{2})$ and $(1, \frac{1}{2})$?
(2) What space do you get if you glue two copies of M along their boundary? *Hint:* Work with $I \times I$, keeping track of equivalence relations. The way that the Möbius boundaries are oriented relative to each other will not matter.

Problem 66. Keep the setup of Problems 64 and 65.

(1) Find triangulations of $I \times I$ and M such that the quotient map from $I \times I$ onto M takes simplices to simplices.
(2) Using the triangulations in (1), verify directly that the Euler characteristic of $I \times I$ is 1, while that of M is 0. Why are these values consistent with Problems 63 and 64?

Problem 67. Suppose that $\mathcal{K}$ is a simplicial complex whose geometric realization $|\mathcal{K}|$ is homeomorphic to $[0, 1]$. Show that there is a finite list of points

$$0 = a_0 < a_1 < \cdots < a_k = 1$$

such that under $|\mathcal{K}| \simeq [0, 1]$, the 1-simplices in $\mathcal{K}$ correspond to the subintervals of the form $[a_i, a_{i+1}]$.

Problem 68. Use a triangulation to compute the mod 2 simplicial homology of the closed interval $[0, 1]$. Using Problem 67, deduce that the answer is the same regardless of the triangulation.

Problem 69. Using the triangulations you found in Problem 62, compute the mod 2 simplicial homology of:

(1) S^1.
(2) D^2.
(3) S^2.

Problem 70. Recall the Klein bottle K from Munkres §75, page 454.

(1) Find a triangulation of K. *Hint:* As with $S^1 \times S^1$, it helps to first describe K as a quoient space of $[0, 1] \times [0, 1]$.
(2) *Hard:* Compute the mod 2 simplicial homology of K and $S^1 \times S^1$. Are they the same or different?
(3) Read in Crossley §9.3 the integral simplicial homology of K and $S^1 \times S^1$. Are they the same or different?

5. Final Exam Topics

The final exam will be a **closed-book** exam. It will be held 12:40–3:00 pm, on Thursday, May 7. It will be at most eleven questions, largely reviewing the content of the previous three exams, plus one or two questions about Section 4.

The best way to prepare for this exam is to try to redo the previous exams without looking at my solutions, then comparing your work against my solutions later. Doing as many of the other practice problems in this file as possible will also help.

Only Topics Included.

- Paths and path connectedness
  - The definitions
  - The definition of $\gamma_2 * \gamma_1$
  - What happens to paths in X under continuous maps $f\colon X \to Y$
  - Why the image of a path connected space is path connected
- Homotopies and path homotopies
  - The definitions and how they differ
  - How to concatenate homotopies (Problem 9)
  - How homotopy, *resp.* path homotopy, defines an equivalence relation on continuous maps, *resp.* paths
- Homotopy equivalences
  - The definition
  - What it means for a space to be contractible
  - How star-convex is different from convex
  - Why star-convex subsets of $\mathbf{R}^N$ are contractible
  - Contractible spaces that are not star-convex
  - Which letters of the alphabet are homotopy equivalent
- The fundamental group $\pi_1(X, x)$
  - The definition
  - What happens if we change the point x
  - What happens when $X \simeq X_1 \times X_2$
  - The basic examples $\mathbf{R}, S^1, S^2, S^1 \times S^1, S^1 \vee S^1$
- The homomorphism $f_*\colon \pi_1(X, x) \to \pi_1(Y, f(x))$ arising from a continuous map $f\colon X \to Y$ and point $x \in X$:
  - The definition
  - What happens with f_* when $f = f_2 \circ f_1$
  - What happens when f is a homotopy equivalence
  - What happens when f is a section
  - What happens when f is a retraction
- Explicit loop representing the elements of $\pi_1(S^1, q)$, where $q = (1, 0)$

- Covering maps
  - The definition
  - Non-examples of covering maps
  - Examples of covering maps of the form $\mathbf{R} \to S^1$ and of the form $S^1 \to S^1$
  - Examples of covering maps to $S^1 \times S^1$ and $S^1 \vee S^1$
  - The path-lifting and path-homotopy lifting properties of covering maps
  - The effect of a covering map on fundamental groups
- Separations of surfaces
  - The definition
  - The definition of a simple closed curve in a space X
  - The Jordan separation theorem: in particular, to which surfaces it does or does not apply
- How to use Problem 39 to show there is no retraction from the closed disk D^2 onto its boundary, the circle S^1
- The free group F_X on a set X
  - The definition
  - Which free groups are isomorphic to the group formed by the integers $\mathbf{Z}$ under addition
  - Examples of the homomorphism $F_X \to G$ arising from an arbitrary map of sets $X \to G$, where G is a group
- Generating sets for a group G
  - The definition
  - Examples where G is $\mathbf{Z}$ or $\mathbf{Z}/m\mathbf{Z}$ or $\mathbf{Z}^2$ or F_X
- Group presentations $\langle s \in S \mid r \in R \rangle$
  - The definition, as a quotient group of F_S
  - Examples of presentations for $\mathbf{Z}, \mathbf{Z}/m\mathbf{Z}, \mathbf{Z}^2$
  - The definition of a free product of groups $G_1 \star G_2$, given presentations for G_1 and G_2
- The Seifert–van Kampen theorem
  - The hypotheses of the theorem
  - The statement
  - How it can be used to solve Problem 28
  - How it lets us compute the fundamental groups of $S^1 \times S^1$, the projective plane P^2 and other "dunce cap" surfaces, and the Klein bottle K
- Simplicial complexes
  - Definition of general position
  - Definition of a simplex
  - Definition of a simplicial complex and its geometric realization
  - Examples and non-examples of simplicial complexes involving a small number of points
  - Examples of triangulations

- Euler characteristic
  - The definition
  - How to compute the Euler characteristic of a graph
  - How to compute the Euler characteristic of (the surface of) a convex polyhedron
  - How Euler characteristic interacts with homotopy equivalences